



NAMPHIMO

Updating National AMPHibian Monitoring in
Luxembourg

NEWTCAM

Camera system for freshwater wildlife
monitoring

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History of National Amphibian monitoring in Luxembourg



IAS

MIE

PNPN3

EU Pledges

Sp. Conservation
Action Plans

Nature Restoration Law

From **species-specific** protocols
dedicated to **Art. 17** needs...

...to the need to **integrate multi-taxa**
and fulfil a **range of obligations**.

From low tech...

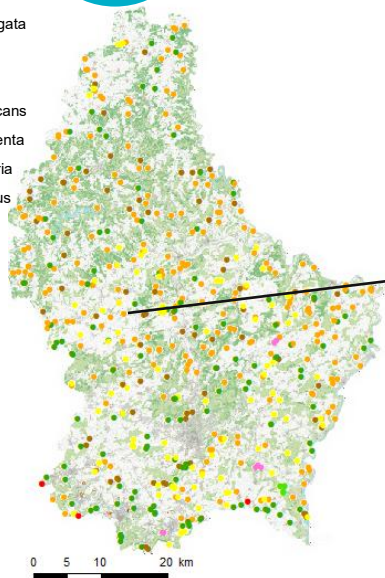


... to high tech

History of National Amphibian monitoring in Luxembourg

2010

- *Bombina variegata*
- *Bufo calamita*
- *Hyla arborea*
- *Alytes obstetricans*
- *Rana kl. esculenta*
- *Rana temporaria*
- *Triturus cristatus*



Ponds existing in 2010-2012 are surveyed on 3-6 year cycle...

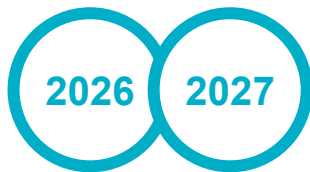
...missing recently created ponds...

...specific protocol missing other species...

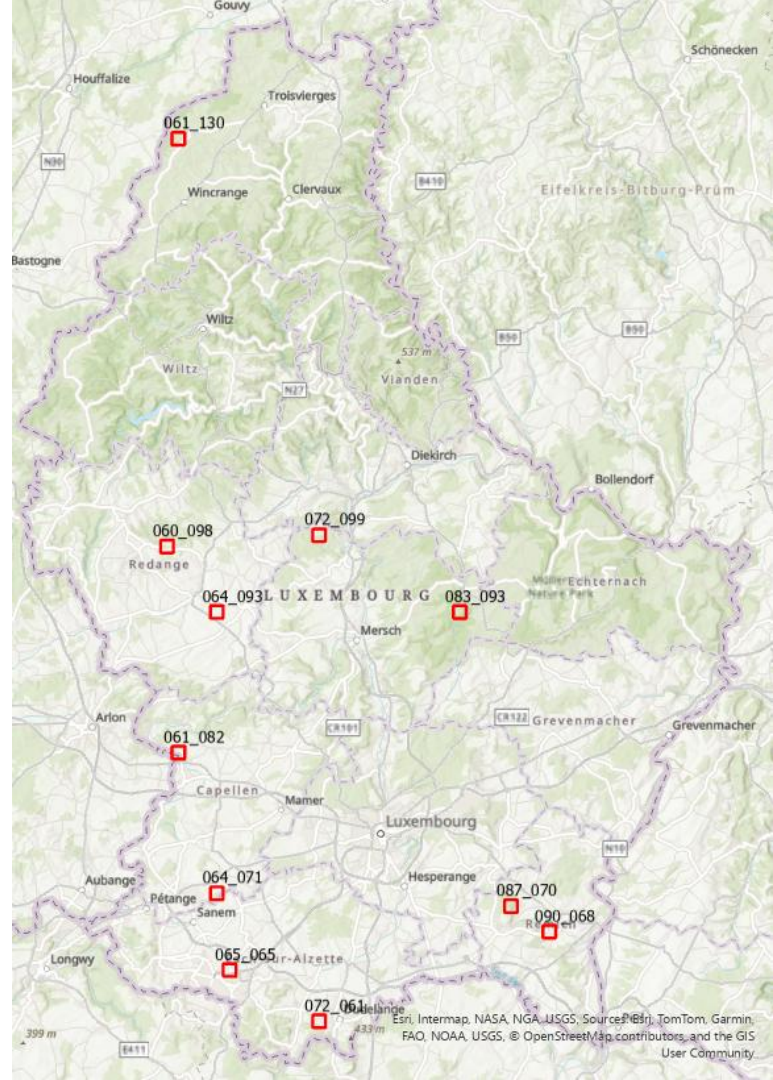
...temporal resolution to detect site occupancy changes...

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The project



- 2 years
- Experimentation in 11 squares (1km)
- Combination of field observation methods at square level
- Beyond amphibian, multi-taxa



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The project



Transect-based survey

ObsMapp Looking for all aquatic habitats
6x/year



Sound recorders

Up to 5 by square
4.5 months



Funnel traps

Up to 5 ponds by square
3x/year



eDNA

Up to 5 ponds by square
3x/year
Amphibian and vertebrates

Field experimentation schedule 2026

	Feb		March				Apr					May					June			Jul	Aug
	s3	s4	s1	s2	s3	s4	s1	s2	s3 (+24h)		s4	s1 (+24h)		s2	s3	s4	s1/s2 (+24h)		s3/s4	s3/s4	s3/s4
	P1		P2		P3	(P4)	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	P16	P17	P18	P19
Transect-based			K0		K1			K2			K3			K4			K5				
Funnel traps									K0			K1					K2				
Bioacoustics					K0										K1					K2	K3
eDNA											K0							K1		K2	
Capt.-Recapt. Bombina														K0			K1			K2	

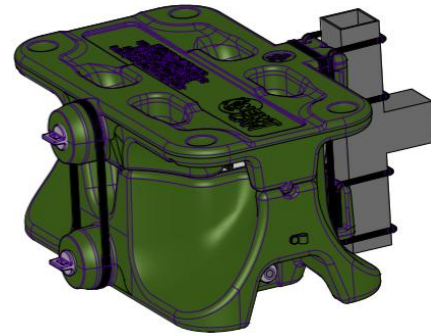
<i>Alytes obsterticans</i>																					
<i>Bombina variegata</i>																					
<i>Hyla arborea</i>																					
<i>Epidalea calamita</i>																					
<i>Pelophylax</i>																					
<i>Triturus cristatus</i>																					
<i>Rana temporaria</i>																					

Field experimentation schedule 2027

	Feb		March				Apr				May				June			Jul	Aug		
	s3	s4	s1	s2	s3	s4	s1	s2	s3 (+24h)		s4	s1 (+24h)		s2	s3	s4	s1/s2 (+24h)		s3/s4	s3/s4	
	P1		P2		P3	(P4)	P5	P6	P7	P8	P9	P10	P11	P12	P13	P14	P15	P16	P17	P18	P19
<i>Transect-based</i>			K0		K1			K2			K3			K4			K5				
<i>Funnel traps</i>									K0			K1					K2				
<i>Bioacoustics</i>																					
<i>eDNA</i>																					
<i>Capt.-Recapt. Bombina</i>														K0			K1			K2	
<i>Capt.-Recapt. NewtCAM</i>			K0		K1		K2		K3		K4		K5	K6	K7	K8	K9				



- ➔ Identify optimal combination to:
- Maximise amphibian detection probabilities and population records
 - Maximise number of species observed (by-catch), especially IAS
 - Minimise monitoring costs



An underwater camera system for freshwater wildlife monitoring

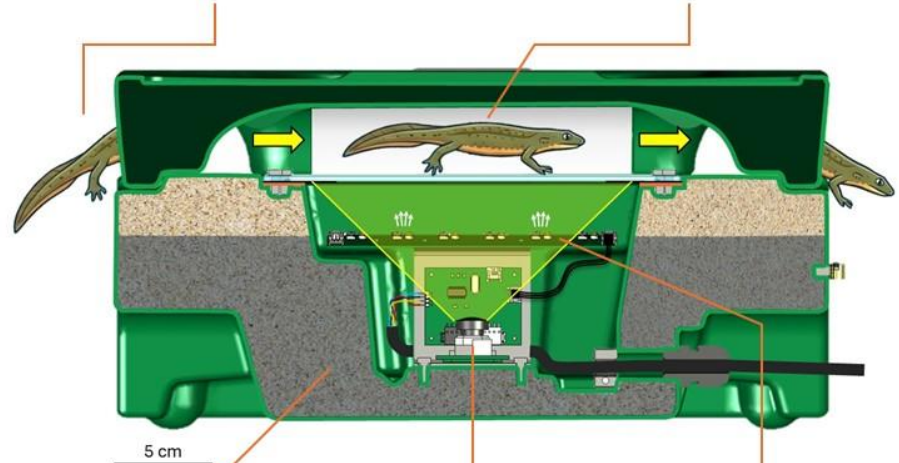


NEWTCAM: an underwater camera trap for freshwater wildlife monitoring



Animals can enter and leave the trap without any stress

Animal movement triggers the recording of images/videos



The ballast system allows the operator to adjust the position and orientation of the trap in the water column

Visible or infrared light can be scheduled to operate continuously or in alternating mode

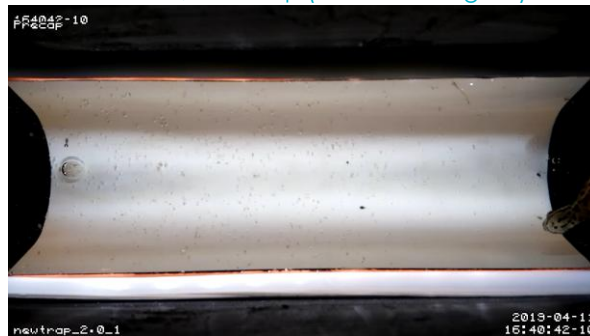
Depending on the orientation, the camera can record ventral, dorsal or lateral images/videos

NEWTFLIX

Palmate newt, male (*Lissotriton helveticus*)



Smooth newt, courtship (*Lissotriton vulgaris*)



Great crested newt predating palmate newt



Alpine newt, female (*Ichthyosaura alpestris*)



Grass snake (*Natrix helvetica*)



Great crested newt and horse leech



NEWTCAM: an underwater camera trap for freshwater wildlife monitoring

Field experimentation: observed species

Ranatra linearis, *Dytiscidae* sp



Nepa cinerea



Pelophylax kl. esculentus



Dytiscidae sp



Great crested newt, larvae



Great crested newt, battle

